O*NET OnLine

Computer Systems Analysts
15-1211.00

Bright Outlook | Updated 2026

Analyze science, engineering, business, and other data processing problems to develop and implement solutions to complex applications problems, system administration issues, or network concerns. Perform systems management and integration functions, improve existing computer systems, and review computer system capabilities, workflow, and schedule limitations. May analyze or recommend commercially available software.

Sample of reported job titles: Applications Analyst, Business Systems Analyst, Computer Analyst, Computer Systems Analyst, Computer Systems Consultant, Information Systems Analyst (ISA), IT Analyst (Information Technology Analyst), IT Systems Analyst (Information Technology Systems Analyst), Programmer Analyst, Systems Analyst

Also see: Health Informatics Specialists

Summary | Details | Custom | Easy Read | Veterans | Español

Contents

Occupation-Specific Information

Tasks

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Importance	Category	Task
81	Core	Troubleshoot program and system malfunctions to restore normal functioning.
76	Core	Provide staff and users with assistance solving computer-related problems, such as malfunctions and program problems.
75	Core	Test, maintain, and monitor computer programs and systems, including coordinating the installation of computer programs and systems.

Importance	Category	Task
74	Core	Use the computer in the analysis and solution of business problems, such as development of integrated production and inventory control and cost analysis systems.
70	Core	Coordinate and link the computer systems within an organization to increase compatibility so that information can be shared.
68	Core	Use object-oriented programming languages, as well as client and server applications development processes and multimedia and Internet technology.
65	Core	Analyze information processing or computation needs and plan and design computer systems, using techniques such as structured analysis, data modeling, and information engineering.
64	Core	Consult with management to ensure agreement on system principles.
62	Core	Specify inputs accessed by the system and plan the distribution and use of the results.
62	Core	Expand or modify system to serve new purposes or improve work flow.

Occupational Requirements

Work Activities

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Importance	Work Activity
99	**Working with Computers** — Using computers and computer systems (including hardware and software) to program, write software, set up functions, enter data, or process information.
90	**Getting Information** — Observing, receiving, and otherwise obtaining information from all relevant sources.
90	**Making Decisions and Solving Problems** — Analyzing information and evaluating results to choose the best solution and solve problems.

Importance	Work Activity
86	**Processing Information** — Compiling, coding, categorizing, calculating, tabulating, auditing, or verifying information or data.
86	**Updating and Using Relevant Knowledge** — Keeping up-to-date technically and applying new knowledge to your job.
79	**Analyzing Data or Information** — Identifying the underlying principles, reasons, or facts of information by breaking down information or data into separate parts.
79	**Communicating with Supervisors, Peers, or Subordinates** — Providing information to supervisors, co-workers, and subordinates by telephone, in written form, e-mail, or in person.
76	**Monitoring Processes, Materials, or Surroundings** — Monitoring and reviewing information from materials, events, or the environment, to detect or assess problems.
70	**Identifying Objects, Actions, and Events** — Identifying information by categorizing, estimating, recognizing differences or similarities, and detecting changes in circumstances or events.
67	**Evaluating Information to Determine Compliance with Standards** — Using relevant information and individual judgment to determine whether events or processes comply with laws, regulations, or standards.

Detailed Work Activities

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- Troubleshoot issues with computer applications or systems.
- Provide technical support for software maintenance or use.
- Coordinate software or hardware installation.
- Monitor computer system performance to ensure proper operation.
- Test software performance.
- Apply information technology to solve business or other applied problems.
- Configure computer networks.
- Write computer programming code.
- Analyze project data to determine specifications or requirements.
- Design integrated computer systems.

Work Context

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E-Mail — How frequently does your job require you to use E-mail?

100% Every day

Telephone Conversations — How often do you have telephone conversations in this job?

79% Every day

21% Once a week or more but not every day

Indoors, Environmentally Controlled — How often does this job require working indoors in an environmentally controlled environment (like a warehouse with air conditioning)?

93% Every day

Determine Tasks, Priorities and Goals — How much freedom does the worker have in determining the tasks, priorities, or goals of the job?

65% A lot of freedom

Importance of Being Exact or Accurate — How important is being very exact or highly accurate in performing this job?

54% Extremely important

45% Very important

Contact With Others — How much does this job require the worker to be in contact with others (face-to-face, by telephone, or otherwise) in order to perform it?

73% Constant contact with others

14% Contact with others most of the time

11% Occasional contact with others

Work With or Contribute to a Work Group or Team — How important is it to work with or contribute to a work group or team in this job?

63% Extremely important

22% Very important

14% Fairly important

Face-to-Face Discussions with Individuals and Within Teams — How frequently does your job require face-to-face discussions with individuals and within teams?

32% Every day

63% Once a week or more but not every day

Freedom to Make Decisions — How much decision making freedom, without supervision, does the job offer?

34% A lot of freedom

58% Some freedom

Spend Time Sitting — How much does this job require sitting?

53% Continually or almost continually

22% More than half the time

14% About half the time

11% Less than half the time

Experience Requirements

Job Zone

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Title	Job Zone Four: Considerable Preparation Needed
Education	Most of these occupations require a four-year bachelor's degree, but some do not.
Related Experience	A considerable amount of work-related skill, knowledge, or experience is needed for these occupations. For example, an accountant must complete four years of college and work for several years in accounting to be considered qualified.
Job Training	Employees in these occupations usually need several years of work-related experience, on-the-job training, and/or vocational training.
Job Zone Examples	Many of these occupations involve coordinating, supervising, managing, or training others. Examples include real estate brokers, sales managers, database administrators, graphic designers, conservation scientists, art directors, and cost estimators.
SVP Range	(7.0 to < 8.0)

Training & Credentials

State training Select a State Go

Local training ZIP Code Go

Certifications Find certifications nationwide

State licenses Select a State Go

Apprenticeship Opportunities

All 5 displayed

Example apprenticeship titles for this occupation:

- Communications-Computer Systems Planning and Implementation(Mil Only)
- Computer Systems Analyst
- Computer Systems Analyst (Nof)
- It Analyst
- Public Health Informatician (Cdc)

Specific title(s) listed above are vetted by industry and approved by the U.S. Department of Labor for use in a Registered Apprenticeship Program.

Start your career and build your skillset. Visit **Apprenticeship.gov** to learn about opportunities related to this occupation.

Worker Requirements

Software Skills

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- **Configuration management software** — Chef; HyperSpace; Perforce Helix software; Puppet
- **Data base management system software** — Amazon DynamoDB; Apache Hive; Oracle Database; Oracle PL/SQL
- **Data base user interface and query software** — Blackboard software; IBM DB2; ServiceNow; Transact-SQL
- **Development environment software** — Apache Kafka; Apache Maven; Oracle Java 2 Platform Enterprise Edition J2EE; Oracle SQL Developer
- **Enterprise application integration software** — Extensible markup language XML; IBM InfoSphere DataStage; Microsoft SQL Server Integration Services SSIS; WebFOCUS
- **Enterprise resource planning ERP software** — Microsoft Dynamics; Oracle PeopleSoft; SAP ERP; SAP software
- **Object or component oriented development software** — C#; jQuery; Scala; Swift
- **Operating system software** — Bash; Microsoft Windows Server; Red Hat Enterprise Linux; UNIX Shell
- **Program testing software** — Functional testing software; Hewlett Packard LoadRunner; JUnit; Selenium
- **Web platform development software** — Django; Google Angular; React; Spring Framework

Hot Technologies are requirements most frequently included across all employer job postings.
See all 119 Hot Technologies for this occupation.

In Demand skills are frequently included in employer job postings for this occupation.
See all 10 In Demand skills for this occupation.

Essential Skills

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Importance	Essential Skill
75	**Speaking** — Talking to others to convey information effectively.
72	**Reading Comprehension** — Understanding written sentences and paragraphs in work-related documents.
69	**Active Listening** — Giving full attention to what other people are saying, taking time to understand the points being made, asking questions as appropriate, and not interrupting at inappropriate times.
69	**Critical Thinking** — Using logic and reasoning to identify the strengths and weaknesses of alternative solutions, conclusions, or approaches to problems.
66	**Active Learning** — Understanding the implications of new information for both current and future problem-solving and decision-making.
60	**Monitoring** — Monitoring/Assessing performance of yourself, other individuals, or organizations to make improvements or take corrective action.
60	**Writing** — Communicating effectively in writing as appropriate for the needs of the audience.
53	**Learning Strategies** — Selecting and using training/instructional methods and procedures appropriate for the situation when learning or teaching new things.
50	**Mathematics** — Using mathematics to solve problems.
31	**Science** — Using scientific rules and methods to solve problems.

Transferable Skills

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Importance	Transferable Skill
69	**Systems Analysis** — Determining how a system should work and how changes in conditions, operations, and the environment will affect outcomes.
69	**Systems Evaluation** — Identifying measures or indicators of system performance and the actions needed to improve or correct performance, relative to the goals of the system.
66	**Judgment and Decision Making** — Considering the relative costs and benefits of potential actions to choose the most appropriate one.
60	**Complex Problem Solving** — Identifying complex problems and reviewing related information to develop and evaluate options and implement solutions.
56	**Programming** — Writing computer programs for various purposes.
53	**Coordination** — Adjusting actions in relation to others' actions.
53	**Social Perceptiveness** — Being aware of others' reactions and understanding why they react as they do.
53	**Time Management** — Managing one's own time and the time of others.
53	**Troubleshooting** — Determining causes of operating errors and deciding what to do about it.
50	**Operations Analysis** — Analyzing needs and product requirements to create a design.

Knowledge

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Importance	Knowledge
91	**Computers and Electronics** — Knowledge of circuit boards, processors, chips, electronic equipment, and computer hardware and software, including applications and programming.
68	**Customer and Personal Service** — Knowledge of principles and processes for providing customer and personal services. This includes customer needs

Importance	Knowledge
	assessment, meeting quality standards for services, and evaluation of customer satisfaction.
62	**English Language** — Knowledge of the structure and content of the English language including the meaning and spelling of words, and rules of composition and grammar.
56	**Mathematics** — Knowledge of arithmetic, algebra, geometry, calculus, statistics, and their applications.
55	**Administration and Management** — Knowledge of business and management principles involved in strategic planning, resource allocation, human resources modeling, leadership technique, production methods, and coordination of people and resources.
50	**Education and Training** — Knowledge of principles and methods for curriculum and training design, teaching and instruction for individuals and groups, and the measurement of training effects.
49	**Telecommunications** — Knowledge of transmission, broadcasting, switching, control, and operation of telecommunications systems.
43	**Engineering and Technology** — Knowledge of the practical application of engineering science and technology. This includes applying principles, techniques, procedures, and equipment to the design and production of various goods and services.
39	**Administrative** — Knowledge of administrative and office procedures and systems such as word processing, managing files and records, stenography and transcription, designing forms, and workplace terminology.
36	**Personnel and Human Resources** — Knowledge of principles and procedures for personnel recruitment, selection, training, compensation and benefits, labor relations and negotiation, and personnel information systems.

Education

How much education does a new hire need to perform a job in this occupation? Respondents said:

- Associate's degree or other 2-year degree required for some jobs
- Bachelor's degree required for some jobs
- Some college, no degree required for some jobs

Worker Characteristics

Abilities

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Importance	Ability
78	**Deductive Reasoning** — The ability to apply general rules to specific problems to produce answers that make sense.
75	**Information Ordering** — The ability to arrange things or actions in a certain order or pattern according to a specific rule or set of rules (e.g., patterns of numbers, letters, words, pictures, mathematical operations).
75	**Oral Comprehension** — The ability to listen to and understand information and ideas presented through spoken words and sentences.
75	**Oral Expression** — The ability to communicate information and ideas in speaking so others will understand.
75	**Written Comprehension** — The ability to read and understand information and ideas presented in writing.
72	**Inductive Reasoning** — The ability to combine pieces of information to form general rules or conclusions (includes finding a relationship among seemingly unrelated events).
72	**Near Vision** — The ability to see details at close range (within a few feet of the observer).
72	**Problem Sensitivity** — The ability to tell when something is wrong or is likely to go wrong. It does not involve solving the problem, only recognizing that there is a problem.
66	**Speech Recognition** — The ability to identify and understand the speech of another person.
63	**Fluency of Ideas** — The ability to come up with a number of ideas about a topic (the number of ideas is important, not their quality, correctness, or creativity).

Career Interest Types

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Occupational Interest	Career Interest Type
90	**Investigative** — Work involves studying and researching non-living objects, living organisms, disease or other forms of impairment, or human behavior. Investigative occupations are often associated with physical, life, medical, or social sciences, and can be found in the fields of humanities, mathematics/statistics, information technology, or health care service.
82	**Conventional** — Work involves following procedures and regulations to organize information or data, typically in a business setting. Conventional occupations are often associated with office work, accounting, mathematics/statistics, information technology, finance, or human resources.
38	**Realistic** — Work involves designing, building, or repairing of equipment, materials, or structures, engaging in physical activity, or working outdoors. Realistic occupations are often associated with engineering, mechanics and electronics, construction, woodworking, transportation, machine operation, agriculture, animal services, physical or manual labor, athletics, or protective services.
21	**Enterprising** — Work involves managing, negotiating, marketing, or selling, typically in a business setting, or leading or advising people in political and legal situations. Enterprising occupations are often associated with business initiatives, sales, marketing/advertising, finance, management/administration, professional advising, public speaking, politics, or law.
16	**Artistic** — Work involves creating original visual artwork, performances, written works, food, or music for a variety of media, or applying artistic principles to the design of various objects and materials. Artistic occupations are often associated with visual arts, applied arts and design, performing arts, music, creative writing, media, or culinary art.
15	**Social** — Work involves helping, teaching, advising, assisting, or providing service to others. Social occupations are often associated with social, health care, personal service, teaching/education, or religious activities.

Specific Interest Areas

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Occupational Interest	Specific Interest Area
93	**Information Technology** — Work involves designing, developing, testing, and maintaining computer software, hardware, networks, and systems.
54	**Mathematics/Statistics** — Work involves developing and applying mathematics and statistical theory and methods.

Work Styles

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Impact	Work Style
86	**Attention to Detail** — A tendency to be detail-oriented, organized, and thorough in completing work.
78	**Intellectual Curiosity** — A tendency to seek out and acquire new work-related knowledge and obtain a deep understanding of work-related subjects.
77	**Innovation** — A tendency to be inventive, to be imaginative, and to adopt new perspectives on ways to accomplish work.
72	**Dependability** — A tendency to be reliable, responsible, and consistent in meeting work-related obligations.
62	**Adaptability** — A tendency to be open to and comfortable with change, new experiences, or ideas at work.
59	**Integrity** — A tendency to be honest and ethical at work.
58	**Perseverance** — A tendency to exhibit determination and resolve to perform or complete tasks in the face of difficult circumstances or obstacles at work.
57	**Cautiousness** — A tendency to be careful, deliberate, and risk-avoidant when making work-related decisions or doing work.
57	**Tolerance for Ambiguity** — A tendency to be comfortable with ambiguity and uncertainty at work.
54	**Initiative** — A tendency to be proactive and take on extra responsibilities and tasks that may fall outside of one's required work role.

Workforce Characteristics

Wages & Employment Trends

Median wages (2025) $50.89 hourly, $105,850 annual

State wages Select a State Go

Local wages ZIP Code Go

Employment (2025) 544,400 employees

Projected growth (2025-2035) ■■■■ Much faster than average (7% or higher)

Projected job openings (2025-2035) 32,900

State trends Select a State Go

Top industries (2025)
Professional, Scientific, and Technical Services (33% employed in this sector)
Finance and Insurance (13%)
Management of Companies and Enterprises (10%)

Source: Bureau of Labor Statistics 2025 wage data and 2025-2035 employment projections. "Projected growth" represents the estimated change in total employment over the projections period (2025-2035). "Projected job openings" represent openings due to growth and replacement.

Job Openings on the Web

State job openings Select a State Go

Local job openings ZIP Code Go

More Information

Related Occupations

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- 11-3021.00 Computer and Information Systems Managers Bright Outlook
- 15-1299.08 Computer Systems Engineers/Architects
- 15-1242.00 Database Administrators

- 15-1243.00 Database Architects
- 15-1211.01 Health Informatics Specialists
- 15-1212.00 Information Security Analysts
- 15-1299.05 Information Security Engineers
- 15-1244.00 Network and Computer Systems Administrators
- 15-1252.00 Software Developers
- 15-1253.00 Software Quality Assurance Analysts and Testers

Professional Associations

Disclaimer: Sources are listed to provide additional information on related jobs, specialties, and/or industries. Links to non-DOL Internet sites are provided for your convenience and do not constitute an endorsement.

National Associations

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- AFCEA International
- Association for Computing Machinery
- Computing Research Association
- IEEE Computer Society
- Institute of Electrical and Electronics Engineers

Accreditation, Certification, & Unions

- CompTIA
- Cyber Degrees
- Institute for Certification of Computing Professionals
- International Institute of Business Analysis
- Project Management Institute